CSC 251 HOMEWORK SEVEN

CHICKWEIGHT EXPLORATORY DATA ANALYSIS

THE ASSIGNMENT

This assignment provides the opportunity to explore a data set using the exploratory data analysis techniques discussed in class, including subsetting, grouping, manipulation, and data visualization. You will investigate patterns within the data, summarize characteristics of the data set, and describe qualitative relationships between the variables.

THE CHICKWEIGHT DATASET

Unlike the coding projects you've been assigned in the past, this assignment is an exploratory exercise with general deliverables. As with a real-world data analysis project, the analyst is looking for patterns, trends, and exceptions in the data without knowing what they may expect. Rather than explicitly directing you to create a plot of variable x and variable y and then describe the relationship, you will be told to "investigate the data" without knowing whether you'll find anything particularly interesting or not. If your analysis of this dataset is sufficient, there are some trends that you should note and observations that you should make.

A poultry farmer from eastern North Carolina has approached you to analyze data for young chicks on his farm that were fed 4 different diets. The data is contained in a file named `ChickWeight.csv` which describes the weights of the young chicks on the different diets at various ages. The weight column provides the body weight of the chick in grams, the Time column gives the number of days since birth when the weight measurement was taken, the Chick column indicates the unique identifier of the chick, and the Diet column indicates which of the four diets the chick received.

EXPECTATIONS

Be sure to include the following elements in your EDA Jupyter notebook:

- Your name, date, and a description of the lab in a Markdown cell at the top of the notebook.
- Installation of any packages your script uses in the first code cell. You should only use packages discussed in class which includes NumPy, pandas, matplotlib, and Seaborn.
- Loading the CSV data file into your notebook.
- Exploration of the dataset using functions discussed in class or included in the acceptable packages. Include the following:

  1) An initial list of 3 questions generated about the data you hope to answer through your exploration. These should be included in a Markdown cell after viewing the DataFrame characteristics but before any visualizations are created. Use these questions to guide the exploration of the data.

2) A comment in a Markdown cell with a bullet-point list that provides an overall description of the data: how many chicks? how many data points per chick? how many chicks were fed each diet? etc. Any patterns in the distribution of the data? Any anomalies or outliers?

3) One additional column added to the DataFrame or a new column added to a subset of the data created to provide insight into the data. For an example, see the "percent" column that we added to the tips DataFrame or the average time per interval that we added to the Woolly Worms notebook. Your new column should make mathematical sense.

4) Grouping by similar characteristics to assist you in your analysis. Recall how we grouped the tip data by party size as an example.

5) Include your best 3 or 4 plots to visualize the data set. Resist the urge to include a dozen or more graphs as including more graphs is **not** better, rather it gives the impression that you don't recognize what's most significant. You should make a minimum of 8 graphs as you explore the data but should **comment out all but the best 3-4** before executing and saving your notebook. A figure containing multiple plots will be counted as 1 graph provided the x and y axes use the same variables for different observations. No more than 4 graphs will be graded so you'll want to select the best ones to avoid too much redundancy. You should include 2-3 different types of graphs, some univariate and some bivariate. All graphs should be titled, labeled, and scaled as appropriate. Include a customization of a plot feature that was not discussed in the class notes. **Add a comment in a Markdown cell regarding this customization so that it can easily be found and credit may be given.**

   In a Markdown cell, describe the variation among the data or note variables that exhibit a qualitative correlation based upon your visualizations. As this is an exploratory data analysis **you should not include predictive analysis nor machine learning methods**.

6) Each code cell should be preceded by a Markdown cell that includes a description of the code in that cell, even if the contents of the cell are commented out so that the cell does not run. This includes the commented-out graphs from item 5) above.

7) Include a comment in a Markdown cell at the end of your notebook with at least 3 observations made through your exploratory data analysis. Number your observations and use complete sentences, correct spelling and appropriate terminology. These observations may or may not be answers to the exploratory questions you listed in #1 but should include answers to at least 2 of your questions.

SUBMISSION

Run your Jupyter notebook before saving it so that all cells are executed, except for those intentionally commented-out, and your best 3-4 visualizations are visible. Points will be deducted if we must run your notebook to grade it! Submit a single Jupyter notebook called `hw7LastnameFirstInitial.ipynb`.